COLLEGE SUBMISSION • B.TECH FINAL YEAR

Green Skilling, Sustainability & AI

Exploring how emerging green skills and artificial intelligence together drive a more sustainable future.

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About Me

Adarsh Mishra

B.Tech Computer Science

Enrollment Number
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Branch / Stream
B.Tech – Computer Science

College
R.N.G Patel Institute of Technology

Semester
8th Semester (Final Year)

A Short Introduction

I am a Third-year B.Tech Computer Science student with a strong interest in Artificial Intelligence, full-stack development, and applied sustainability. Over the 1 year, I have worked on AI-driven projects ranging from geospatial environmental monitoring to multi-agent systems, which sparked my curiosity about how technology can directly support the Sustainable Development Goals.

This presentation reflects what I learned while studying green skilling, sustainability practices, and AI's growing role in solving environmental challenges — and how I plan to apply these ideas in my own capstone project.

AI & Machine Learning | Sustainability | Geospatial Tech | Full-Stack Dev

KEY TAKEAWAYS

Things I Learned

Sustainable Development Goals

Studied all 17 SDGs in depth, with special focus on SDG 17 — Partnerships for the Goals.

Green Data Centres

How data centres are redesigning cooling, power, and infrastructure to cut emissions.

Government Initiatives

Explored national programs driving renewable energy, smart cities, and Digital India.

Renewable Energy

Learned how solar, wind, and hydro power are replacing fossil-fuel dependency.

AI for the Environment

Discovered real applications of AI in climate prediction, monitoring, and conservation.

DEEP DIVE

My Favorite Topic — SDG 17: Partnerships for the Goals

17

Goal 17 of 17 — the enabler that connects every other SDG

Why It Matters

SDG 17 recognizes that no single government, company, or individual can solve climate change alone — global goals need global cooperation, shared funding, and technology transfer.

Collaboration in Action

It encourages partnerships between governments, private industry, NGOs, and academic institutions to pool resources, expertise, and innovation toward common sustainability targets.

How AI Helps Achieve It

AI enables large-scale data sharing, cross-border climate monitoring, predictive analytics for policy-making, and open-source tools that let smaller organizations participate in global sustainability efforts.

CASE STUDY

Learning from Data Centre Documentaries

Why Data Centres Consume So Much Energy

Thousands of servers run 24/7 for computing, storage, and networking — and an almost equal amount of energy is spent just on cooling them.

Green Data Centres

Facilities designed with energy-efficient hardware, better airflow design, and sustainable building materials to shrink their footprint.

Renewable Energy Usage

Leading providers now power server farms with solar and wind energy, moving away from fossil-fuel grids.

AI-Based Cooling & Power Optimization

Machine learning models predict cooling needs in real time, cutting energy use significantly, as pioneered by companies like Google DeepMind.

Carbon Emission Reduction

Combining efficient hardware, renewables, and AI optimization helps data centres lower their overall carbon emissions year over year.

Learning from Government Websites

Renewable Energy Push

National missions promoting solar parks, wind corridors, and hydrogen energy to reduce fossil fuel dependence.

Digital India

A nationwide initiative digitizing governance and services, reducing paper use and enabling remote, low-carbon access to public services.

Smart Cities Mission

Integrating IoT and AI-based traffic, waste, and energy management into urban infrastructure for efficient, sustainable cities.

Environmental Conservation

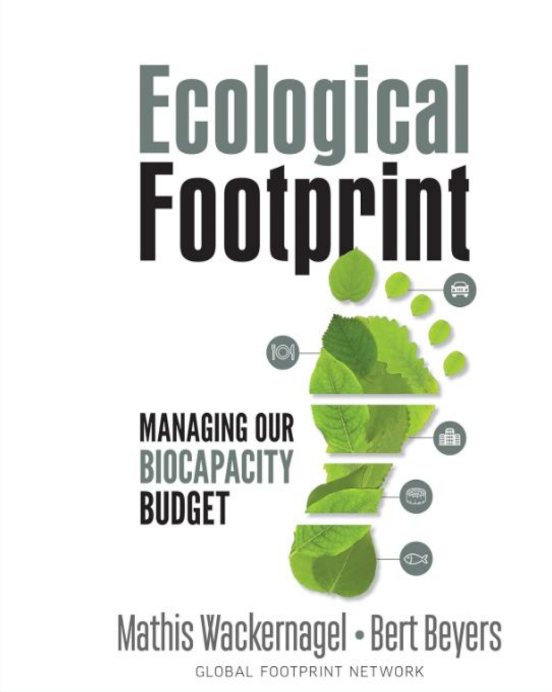
Afforestation drives, river-cleaning programs, and biodiversity protection policies aimed at long-term ecological balance.

PRACTICAL WORK

Activity Screenshots

SDG Learning Module

Government Portal Research



Data Centre Documentary

AI Sustainability Tool

CONCEPT

What is Green Skilling?

Definition

Green skilling is the process of learning skills, knowledge, and technologies needed for jobs that support environmental sustainability — from renewable energy systems to eco-friendly manufacturing.

Why It Matters

As industries shift toward sustainable practices, green skills are becoming essential for employability, opening fast-growing career paths across energy, agriculture, and technology sectors.

Career Opportunities & Examples

- Renewable Energy
- Electric Vehicles
- Waste Management
- Sustainable Agriculture
- AI Environmental Solutions

CONCEPT

What is Sustainability?

Meeting the needs of the present without compromising the ability of future generations to meet their own needs.

Environmental

Protecting ecosystems, biodiversity, air and water quality, and natural resources for the long term.

Social

Ensuring equity, health, education, and quality of life for communities across generations.

Economic

Building growth models that create prosperity without depleting resources or harming the planet.

Everyday Sustainable Practices

Reducing single-use plastic • Saving energy and water at home • Choosing public transport or EVs • Composting and recycling waste • Supporting local, sustainable products

BALANCED VIEW

AI — Benefits and Challenges for Sustainability

Benefits

Smart Agriculture
Precision farming boosts yield while cutting water and pesticide use.

Climate Prediction
ML models forecast weather and climate risks with growing accuracy.

Waste Segregation
Computer vision automates sorting of recyclables at scale.

Energy Optimization
AI balances power grids and reduces overall energy waste.

Water Management
Predictive systems detect leaks and optimize irrigation.

Forest Monitoring
Satellite AI tracks deforestation and wildlife in near real time.

Challenges

High Energy Consumption
Training large AI models requires substantial electricity.

Data Centre Emissions
Growing AI workloads increase the carbon footprint of hosting.

Electronic Waste
Frequent hardware upgrades for AI infrastructure add to e-waste.

Cooling Water Usage
Large data centres consume significant water for cooling systems.

Need for Responsible AI
Governance and ethics must guide sustainable AI deployment.

CAPSTONE PROJECT

EcoVision AI

Intelligent Environmental Monitoring System

Problem Statement

Environmental degradation from air pollution, poor waste segregation, and untracked carbon emissions often goes undetected until damage is severe — real-time, accessible monitoring is missing at the community level.

Objectives

- Provide real-time environmental data to communities
- Automate waste and pollution classification with AI
- Support data-driven policy and personal decisions

Technologies Used

- Python
- Machine Learning
- Computer Vision
- IoT
- Cloud

Expected Impact

Supports SDG 3, 11, 13 & 17 by making environmental data actionable, improving public health awareness, and strengthening community-level climate resilience.

Key Features

Air Quality Monitoring

Real-time AQI tracking via IoT sensors and dashboards.

Waste Classification

Computer vision sorts recyclable vs. non-recyclable waste.

Pollution Prediction

ML forecasts pollution spikes from historical & weather data.

Carbon Footprint Dashboard

Visualizes personal or community emission trends.

AI Chatbot

Answers sustainability questions with personalized eco-tips.

Thank You

Questions & Discussion

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